

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1084

Stonewall Energy Center 138 kV
August 29, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1084. The Interconnection Customer proposes a 179.4 MW wind facility interconnecting at the Stonewall Energy Center 138 kV substation in the Virginia area. The request was received on October 21, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 310 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,452,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: STONEWALL ENERGY CTR-FOUNTAIN 138kV
Contingency: CLARK COUNTY-KEARNEYVILLE 345kV line
Loading: 142.6% (Rating: 223 MVA)
DFAX: 66.3%
Pre-project loading: 89.1%
2. Facility: CHAFFEE GAS RECOVERY-MICHIGAN WIND 2 138kV
Contingency: STONEWALL ENERGY CENTER-FOUNTAIN 138kV L
Loading: 120.1% (Rating: 193 MVA)
DFAX: 70.8%
Pre-project loading: 54.4%
3. Facility: W STAUNTON SUBSTATION-WOODS DP SUBSTATION 138kV
Contingency: FUNTAIN-STONEWALL ENERGY CENTER 138kV
Loading: 117.5% (Rating: 324 MVA)
DFAX: 28.8%
Pre-project loading: 101.5%
4. Facility: DEAN DRIVE AREA ELECTRIC GENERATORS-W STAUNTON 138kV L
Contingency: FOUNTAIN-STONEWALL ENERGY CTR 138kV
Loading: 110.4% (Rating: 244 MVA)
DFAX: 37.2%
Pre-project loading: 83.1%
5. Facility: NOVEC PARADISE-DEAN DRIVE AREA ELECTRIC GENERATORS 138kV
Contingency: STONEWALL ENERGY CTR-FOUNTAIN 138kV line
Loading: 110.4% (Rating: 244 MVA)
DFAX: 37.2%
Pre-project loading: 83.1%

6. Facility: MICHIAN WIND 2-GAINESVILLE 138kV
 Contingency: STONEWALL ENERGY CTR-FOUNTAIN 138kV
 Loading: 101.8% (Rating: 206 MVA)
 DFAX: 70.8%
 Pre-project loading: 40.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STONEWALL ENERGY CTR-FOUNTAIN 138kV	Rebuild line with higher-capacity conductor	\$1,630,000	100% (\$1,630,000)
MICHIGAN WIND 2-CHAFFEE GAS RECOVERY 138kV	Rebuild line with higher-capacity conductor	\$1,084,000	100% (\$1,084,000)
W STAUNTON-WOODS DP 138kV	Rebuild line with higher-capacity conductor	\$803,000	53% (\$426,000)
DEAN DRIVE AREA ELECTRIC GENERATORS-W STAUNTON 138kV L	Rebuild line with higher-capacity conductor	\$582,000	99% (\$575,000)
DEAN DRIVE AREA ELECTRIC GENERATORS-NOVEC PARADISE 138kV line	Rebuild line with higher-capacity conductor	\$2,266,000	99% (\$2,236,000)
MICHIAN WIND 2-GAINESVILLE 138kV	Reconductor line	\$501,000	100% (\$501,000)
	Total allocated to project		\$6,452,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.